Mathematics for Biomedical Engineering

L9:

Differential Equations

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Key Learning Objectives

- To appreciate the importance of differential equations in engineering;
- To understand the general form of a differential equation and the difference between ordinary and partial differential equations;
- To understand the difference between general and particular solutions of differential equations;
- To solve first-order differential equations using separation of variables and integrating factor;
- To learn basic computational approaches to differential equations.

Basic Concepts

Solution of Biomedical Engineering problems often involves dealing with relations of the form

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} = f(t, y(t))$$

For example, $y(t)$ may represent distance and, hence, $\frac{\mathrm{d}y(t)}{\mathrm{d}t}$ is the rate of change of distance with time, t; that is speed.

The equation above is called a **differential equation** because it contains the derivative of an unknown function, $y(t)$.

To **solve a differential equation** means to find the function $y(t)$ satisfying the equation. The **solution** may be **analytical** (the answer can be written down in terms of elementary functions such as e^t, $\sin t$, etc) or **obtained by numerical methods** which produce approximate solutions.

For example, a very simple differential equation is

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} = t^2$$

Ordinary and Partial Differential Equations

A differential equation is any equation which contains derivatives of an unknown function, either ordinary derivatives or partial derivatives.

Ordinary Differential Equations (ODEs) contain an unknown function of **1 variable**

$$m\frac{\mathrm{d}^2\boldsymbol{r}}{\mathrm{d}t^2} = \boldsymbol{F}\left(t, \boldsymbol{r}, \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right) \qquad \text{Newton's Second Law}$$

Partial Differential Equations (PDEs) contain an unknown function of **multiple variables**

$$\frac{\partial T}{\partial t} = c\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) \qquad \text{3-D Heat Equation}$$

This topic focuses on ODEs.

General Form of an ODE

The general form of an ODE can be written as

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + f\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^{n-1}y}{\mathrm{d}x^{n-1}}\right) = 0$$

The **dependent variable**, y, is the variable being differentiated. The variable x is the **independent variable**, since the value of y depends upon the choice we have made for x.

The **order** n of an ODE is the order of its highest derivative.

Most basic ODE: $n = 1$ and $f = f(x)$. Solution: $y = -\int f(x)\,\mathrm{d}x$.

`Example:` State the dependent and independent variables, and the order, of the following differential equations:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x + t^2, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} = y + \sin x, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^3 = x^7$$

Linear ODEs

An ODE is said to be **linear** if (i) the dependent variable and its derivatives occur to the first power only; (ii) there are no products involving the dependent variable and/or its derivatives; (iii) there are no nonlinear functions of the dependent variable, such as sine, exponential, etc. An ODE that is not linear is said to be **nonlinear**.

The linearity of an ODE is not determined or affected by the presence of nonlinear terms involving the independent variable.

The general form of a **linear ODE** is

$$\sum_{j=0}^{n} p_j(x)\frac{\mathrm{d}^j y}{\mathrm{d}x^j} + g(x) = 0,$$

with $p_j(x)$ and $g(x)$ functions of x. They can be classified into:

- **Homogeneous**: $g(x) = 0$;
- **Non-homogeneous**: $g(x) \neq 0$;

`Example:` Decide whether or not the following equations are linear:

$$\sin x\frac{\mathrm{d}y}{\mathrm{d}x} + y = x, \qquad \sin y\frac{\mathrm{d}y}{\mathrm{d}x} + y = x, \qquad \sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \sin y = 0$$

Linear ODEs with Constant Coefficients

A **linear ODE with constant coefficients** is of the form

$$\sum_{j=0}^{n} a_j \frac{\mathrm{d}^j y}{\mathrm{d}x^j} + g(x) = 0,$$

with a_j constant coefficients and $g(x)$ a function of x.

`Example:` Decide whether or not the following equations are linear with constant coefficients:

$$5\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 8y = x^2, \qquad x\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 7\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

The Solution of an ODE

$y(x)$ is a **solution** of the ODE

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + f\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^{n-1}y}{\mathrm{d}x^{n-1}}\right) = 0$$

if $y(x)$ is n times differentiable and satisfies the ODE for all x.

Thus, a solution is found when we have obtained an expression for y in terms of x that can be substituted into the ODE to make the left hand side equal to zero.

`Example:` Show that $y = 5e^{2x}$ is a solution of the equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = 2y$$

There are many different expressions that can satisfy an ODE: that is, there are many solutions.
A solution from which all possible solutions can be found is called a **general solution**.

Conditions and Particular Solutions

The general solution of the equation $\frac{\mathrm{d}y}{\mathrm{d}x} = y$ is $y = Ce^x$, where C is any constant.

C is called an **arbitrary constant**.

By choosing different values of C, different solutions are obtained. Every solution of the ODE can be obtained from this **general solution**.

To determine a particular value for the constant C we need to be given more information in the form of a **condition**.
For example, if we are told that at $x = 0$, $y = 4$, then from $y = Ce^x$ we have

$$4 = Ce^0 = C, \qquad\qquad C = 4.$$

Therefore, $y = 4e^x$ is the solution of the ODE that additionally satisfies the condition $y(0) = 4$. This is called a **particular solution**.

A condition that is to be satisfied at the leftmost point of the interval of interest is called an **initial condition**.

The problem of finding a solution of the differential equation that satisfies the initial condition is called an **initial value problem**.

Solution of an Initial Value Problem

In general, $y(x)$ is a **solution** of the initial value problem

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + f\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}, \ldots, \frac{\mathrm{d}^{n-1}y}{\mathrm{d}x^{n-1}}\right) = 0,$$

if

1. $y(x)$ is n times differentiable and satisfies the ODE for all x in some open interval (a, b) that contains x_0,
2. $y(x)$ satisfies the initial conditions

$$y(x_0) = k_0, \quad y_0'(x_0) = k_1, \ldots, y^{(n-1)}(x_0) = k_{n-1}.$$

The largest open interval that contains x_0 on which y is defined and satisfies the differential equation is the **interval of validity** of y.

`Example:` Show that $x(t) = 2e^{3t}$ is a solution of the initial value problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} - 2x = 2e^{3t}, \qquad x(0) = 2.$$

First order ODEs

In this topic we will focus on first-order ODEs. These have the general form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + f(x, y) = 0$$

We will learn the following techniques commonly used to solve first-order ODEs:

- Separation of variables (for separable **nonlinear** equations)
- Integrating factor (for **linear** non-homogeneous equations)

Separation of Variables

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The technique can be applied to first-order ODEs that can be written in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x)g(y)$$

First-order ODEs that can be written in this way are called **separable**.

The idea is to **rearrange the first-order ODE** to be solved in such a way that all terms involving the dependent variable appear on one side of the equation, and all terms involving the independent variable appear on the other.

Then it is a matter of **integration to complete the solution**.

Separation of Variables

The solution of the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x)g(y)$$

is found from

$$\int \frac{1}{g(y)}\mathrm{d}y = \int f(x)\mathrm{d}x$$

`Example:` Solve the equation $\frac{\mathrm{d}y}{\mathrm{d}x} = 3ye^{x}$ by separation of variables.

Bernoulli Equation

Some equations that do not appear to be separable can be made so by means of a suitable substitution.

A classic example is the nonlinear **Bernoulli equation**, which has the general form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + p(x)y = f(x)y^r,$$

where r can be any real number other than 0 or 1 (for which the equation is linear).

We can transform it into a separable equation by taking $y(x) = u(x)y_1(x)$, with $y_1(x)$ a nontrivial solution of the homogeneous equation.
Substitution of $y(x) = u(x)y_1(x)$ into the original equation yields

$$\frac{\mathrm{d}u}{\mathrm{d}x} = u(x)^r f(x)(y_1(x))^{r-1},$$

which is separable.

Integrating Factor

Key Point

All first-order linear ODEs can be written in standard form as

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

`Example:` Write the equation $L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri = E\sin(\omega t)$ in standard form.

Equations of this type can always be solved by following these steps:

1. Multiply by a function $\mu(x)$ known as **integrating factor**;
2. Integrate both sides;
3. Divide by $\mu(x)$.

Finding an Integrating Factor

When multiplying

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

by the integrating factor $\mu(x)$ we obtain

$$\mu\frac{\mathrm{d}y}{\mathrm{d}x} + \mu P(x)y = \mu Q(x)$$

We will choose $\mu(x)$ so that the left-hand side can be written as $\frac{\mathrm{d}}{\mathrm{d}x}(\mu y)$ and, hence, the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \mu Q(x)$$

It follows immediately by integrating both sides that

$$\mu y = \int \mu Q(x)\mathrm{d}x$$

Dividing by μ we can obtain y.

Choosing $\mu(x)$

We want $\mu(x)$ to satisfy

$$\mu\frac{\mathrm{d}y}{\mathrm{d}x} + \mu P(x)y = \frac{\mathrm{d}}{\mathrm{d}x}(\mu y)$$

Expanding the right-hand side using the product rule for differentiation:

$$\mu\frac{\mathrm{d}y}{\mathrm{d}x} + \mu P(x)y = \mu\frac{\mathrm{d}y}{\mathrm{d}x} + y\frac{\mathrm{d}\mu}{\mathrm{d}x}$$

Comparing the coefficients of y on both sides shows that we must choose μ to satisfy

$$\mu P(x) = \frac{\mathrm{d}\mu}{\mathrm{d}x}$$

This is a first-order ODE that can be solved by separation of variables to obtain

$$\mu = e^{\int P(x)\mathrm{d}x}$$

`Example:` Solve the equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = xe^{-2x}$$

Computational Approaches

The solution of ODEs falls into two categories:

1. **Symbolic solutions**
2. **Numerical solutions**

Symbolic solutions can be obtained using **pen and paper** (e.g. using the algebraic techniques seen in previous slides) and **computer software packages**. For example, the *Symbolic Math Toolbox* of Matlab enables solution of ODEs symbolically.

However, for many ODEs it is impossible to obtain a symbolic answer in terms of common functions.

Techniques known as **numerical methods** exist for obtaining approximate solutions. They are often laborious to perform by hand and so are best implemented using a computer.

In this lecture we will learn the following technique for solving first-order ODEs numerically: **Euler's method**.

But first, an example on how to solve symbolically a first-order ODE using Matlab.

Example of Solving Symbolically a First-Order ODE Using Matlab

The Matlab function *dsolve* computes symbolic solutions to ODEs.

`Example:` Solve the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - x^2 y = 0, \qquad y(0) = 3$$

The command

```
dsolve('Dy-x*x*y=0', 'x')
```

finds the general solution:

`C1*exp(x^3/3)`

The command

```
dsolve('Dy-x*x*y=0', 'y(0)=3', 'x')
```

finds the particular solution:

`3*exp(x^3/3)`

Note that the symbol `D` is used to denote the derivative of the dependent variable with respect to the independent variable.

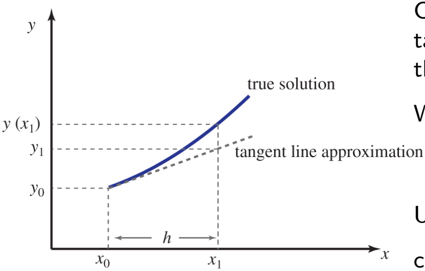
Euler's Method

It is a numerical method for finding approximate solutions of ODEs having the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x,y), \qquad y(x_0) = y_0,$$

When solving this equation numerically we use the fact that $\frac{\mathrm{d}y}{\mathrm{d}x}$ is the gradient of the tangent to $y(x)$.

We know from the given equation that the point (x_0, y_0) lies on the graph of the solution.



Consider moving to a nearby point, (x_1, y_1), on the tangent line, where $x_1 = x_0 + h$. Then (x_1, y_1) is on the tangent line, but not on the solution curve.

We can use y_1 as an approximation to $y(x_1)$:

$$y_1 = y_0 + hf(x_0, y_0)$$

Using the point (x_1, y_1) as a new starting point, we can find an approximate solution, y_2, at $x_2 = x_1 + h$:

$$y_2 = y_1 + hf(x_1, y_1)$$

Euler's Method

Key Point

An approximate solution of

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y),$$

subject to the initial condition $y(x_0) = y_0$, is given by

$$y_{n+1} = y_n + hf(x_n, y_n), \qquad y_0 = y(x_0),$$

where h is the step-size, or **increment**, in x.

`Example:` Use three steps of Euler's method to obtain the approximate solutions of

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{y}, \qquad y(1) = 2$$

shown in the table.

x_n	y_n	true: $y(x_n)$ (to 6 d.p.)
$x_0 = 1$	$y_0 = 2$	2
$x_1 = 1.1$	$y_1 = 2.05$	2.051828
$x_2 = 1.2$	$y_2 = 2.103659$	2.107131
$x_3 = 1.3$	$y_3 = 2.160702$	2.165641

What We Learnt in Topic 9

- To appreciate the importance of differential equations in engineering;
- To understand the general form of a differential equation and the difference between ordinary and partial differential equations;
- To understand the difference between general and particular solutions of differential equations;
- To solve first-order differential equations using separation of variables and integrating factor;
- To learn basic computational approaches to differential equations.